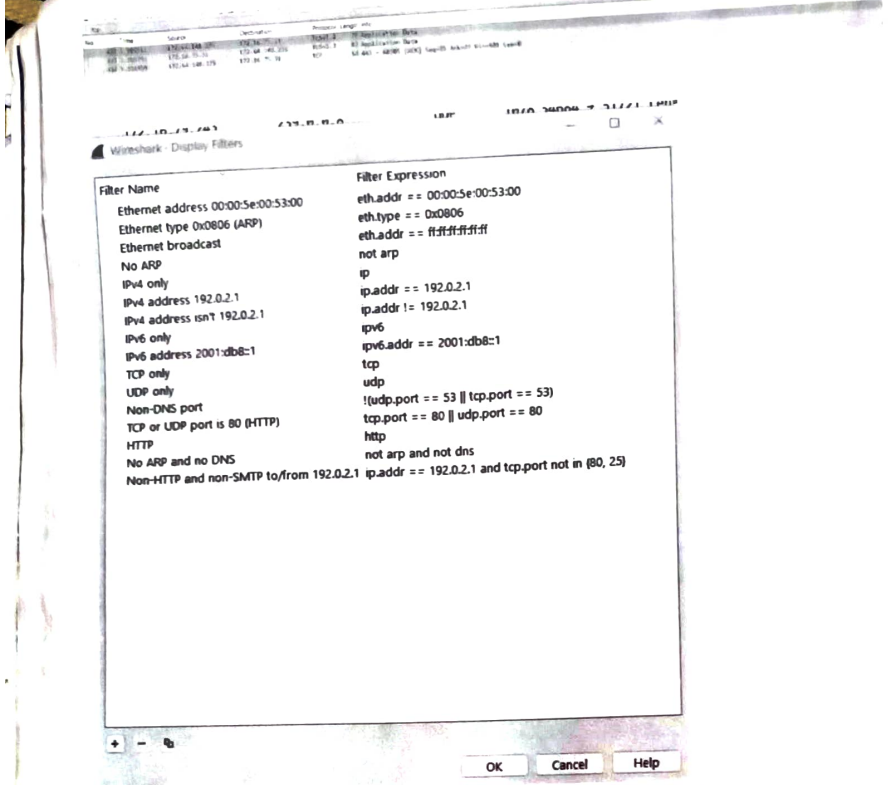
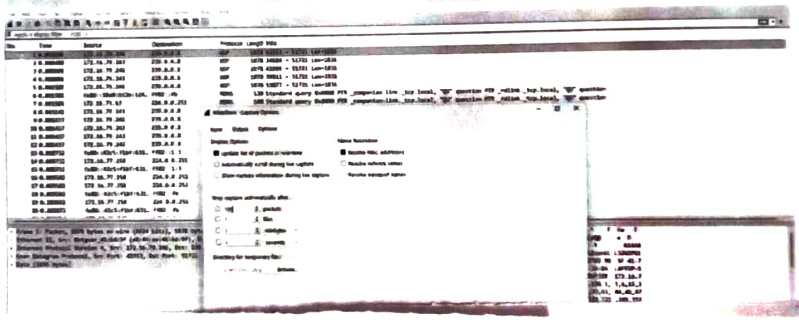
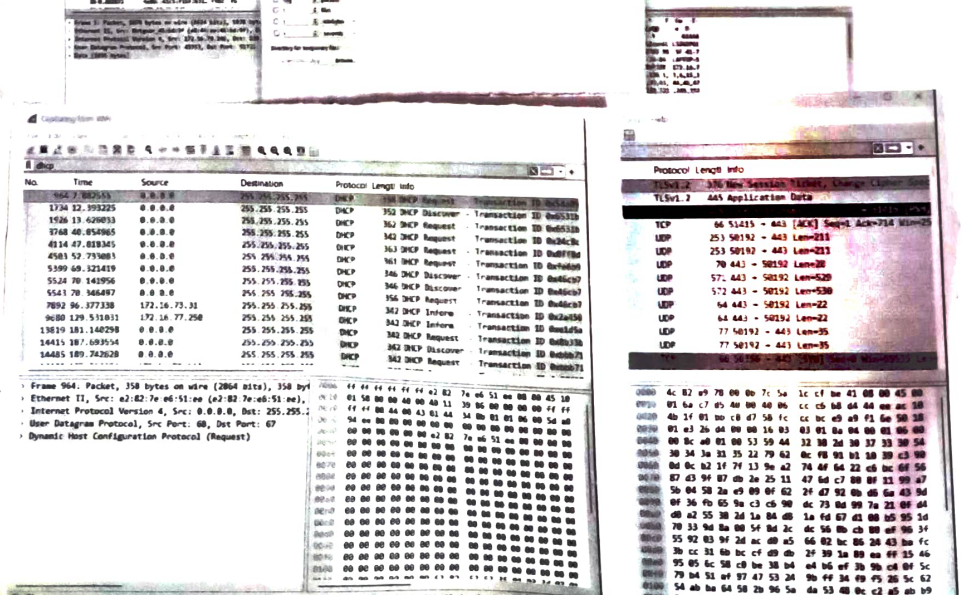






Exp No: 5

Date: 30/1/26 EXPERIMENT ON PACKET CAPTURE : WIRESHARK.

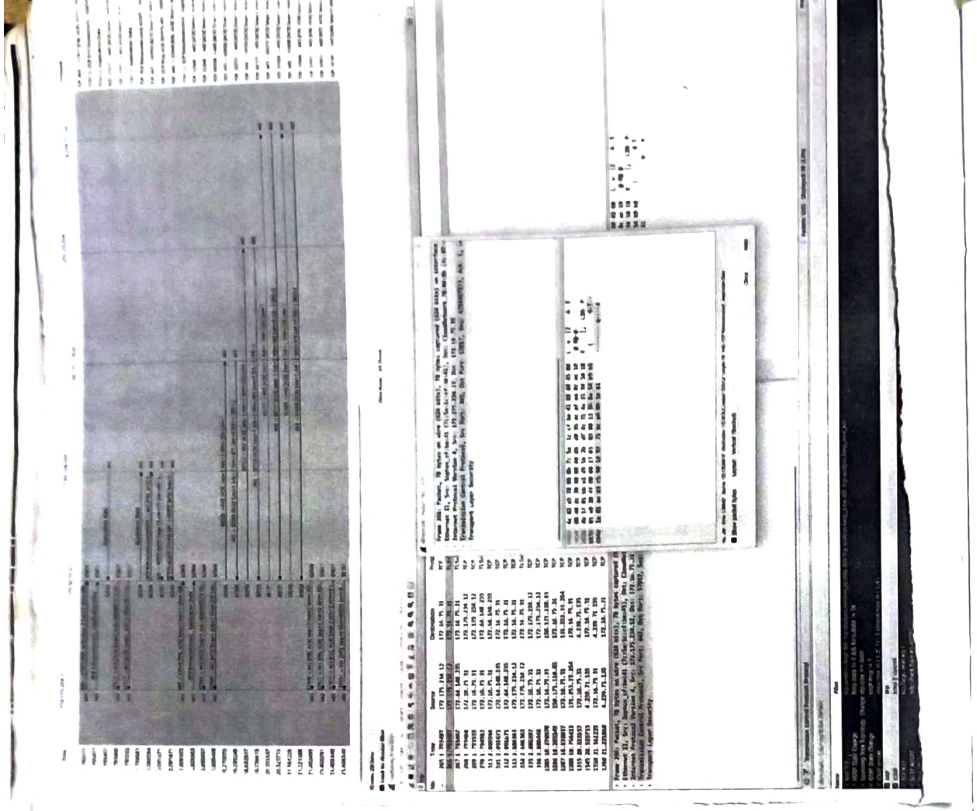
Aim: To perform a experiment on packet capture using wireshark tool.

Procedure:-

1) Create a filter to display only TCP | USP packets, inspects of the packets and provides the flow graphs
   * Select LAN in wireshark.
   * Go to Capture option.
   * Select Stop capture automatically after 100 packets.
   * Then click Start Capture.
   * Search TCP packets in Search bar.
   * To see flow graph, click Statistics flow graph
   * Save the packets.

2) Create a filter to display only ARP packets:
   * Go to Capture and Select Stop Capture.
   * Then click Start Capture and Search ARP
   * Save the packets.

3) Create and display to filter only DNS packets:
   * Go to Capture and Select Stop Capture.
   * Then click Start Capture and Search DNS.
   * To see the flow graph, Click Statistics.
   * Save the packets.



dns

No.	Time	Source	Destination	Protocol	Length	Info
5047	64.823837	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xf2bf HTTPS dns.google
5048	64.823968	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x2c40 A dns.google
5092	64.872708	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x2c40 A dns.go
5093	64.872708	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xf2bf HTTPS dn
5907	73.507194	172.16.75.31	172.16.72.1	DNS	70	Standard query 0xdd84 HTTPS dns.google
5908	73.507610	172.16.75.31	172.16.72.1	DNS	70	Standard query 0x7b95 A dns.google
5909	73.522819	172.16.72.1	172.16.75.31	DNS	102	Standard query response 0x7b95 A dns.go
5910	73.522819	172.16.72.1	172.16.75.31	DNS	70	Standard query response 0xdd84 HTTPS dn
6172	75.347431	172.16.75.31	8.8.8.8	DNS	70	Standard query 0x7402 HTTPS dns.google
6173	75.347571	172.16.75.31	8.8.8.8	DNS	70	Standard query 0xddfb A dns.google
6174	75.377779	8.8.8.8	172.16.75.31	DNS	102	Standard query response 0xddfb A dns.go
6180	75.398557	8.8.8.8	172.16.75.31	DNS	146	Standard query response 0x7402 HTTPS dn

```
> Frame 5047: Packet, 70 bytes on wire (560 bits), 70 byte
> Ethernet II, Src: CloudNetwork_78:00:0b (4c:82:a9:78:00:
> Internet Protocol Version 4, Src: 172.16.75.31, Dst: 172.
> User Datagram Protocol, Src Port: 53918, Dst Port: 53
> Domain Name System (query)
```

```
0000  7c 5a 1c cf be 41 4c 82  a9 78 00 0b 08 00 45 00
0010  00 38 65 85 00 00 80 11  e9 ee ac 10 4b 1f ac 10
0020  48 01 d2 9e 00 35 00 24  99 73 f2 bf 01 00 00 01
0030  00 00 00 00 00 00 03 64  6e 73 06 67 6f 6f 67 6c
0040  65 00 00 41 00 01
```

arp

No.	Time	Source	Destination	Protocol	Length	Info
1343	10.957901	CloudNetwork_78:00:...	Broadcast	ARP	42	Who has 172.16.75.125? Tell 172.16.75.6
1977	14.358615	CloudNetwork_78:00:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2098	15.471882	CloudNetwork_78:00:...	Broadcast	ARP	42	Who has 172.16.77.18? Tell 172.16.75.61
2400	17.952584	CloudNetwork_78:00:...	Broadcast	ARP	42	Who has 172.16.78.5? Tell 172.16.75.61
2747	24.883561	CloudNetwork_78:00:...	Broadcast	ARP	42	Who has 172.16.75.204? Tell 172.16.75.6
4906	61.332363	CloudNetwork_77:ff:...	Broadcast	ARP	42	Who has 172.16.77.93? Tell 172.16.75.53

4). Create a filter to display only HTTP | Ip | ICMP | DHCP and ~~Input~~ Inspect the packets :-

* Select LAN in Wireshark.
* Go to the Capture Option
* Select Stop Capture automatically after 100....
* Then click Start Capture.
* Search HTTP or Ip | ICMP or DHCP in the Searchbar.
* Save the packets.

5. Analysing Packets using Wireshark Tool.

* Create a filter to display only TCP | UDP packets.

* Uses of Networks:
  - Captures networks traffic.
  - Analyzes protocols.

* Application :
  - Network troubleshooting.
  - Security analysis
  - Protocol debugging.

Result:

Thus, the experiment on Capturing packets using wireshark tool has been Successfully implemented.